Hema Sagar Koppusetti

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Professional Summary

Ambitious AI/ML student with strong expertise in data-driven problem solving and machine learning techniques. Proven ability to design and implement intelligent systems, combined with a continuous learning mindset and a focus on impactful solutions.

Education

Lendi Institute of Engineering and Technology — Vizianagaram, Andhra Pradesh
B.Tech in Computer Science and Engineering(AI and ML) – CGPA: 8.79/10.0 — *September 2023 – May 2027*

- Relevant Coursework: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Statistical ML

Smt.Godavari Devi Saraf Senior Secondary School — Garividi, Andhra Pradesh
XII (CBSE) – Percentage: 85.0% — *June 2021 – March 2023*

Smt.Godavari Devi Saraf Senior Secondary School — Garividi, Andhra Pradesh
X (CBSE) – Percentage: 90.0% — *April 2020 – March 2021*

Technical Skills

Programming Languages: Python, C, Java, SQL
ML Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM
Deep Learning: CNN, RNN, LSTM, GAN, Transformer, BERT, GPT, Transfer Learning
Cloud Platforms: AWS, Google Cloud (Vertex AI)
Tools & Libraries: Git, Jupyter, NumPy, Pandas, OpenCV, Hugging Face
Specializations: Computer Vision, NLP, Reinforcement Learning, Generative AI

Projects

Self-Supervised Learning for Robust Image Understanding| Project Link
PyTorch, Vision Transformers, Contrastive Learning — March 2025 – April 2025

- Implemented a SimCLR-based contrastive self-supervised learning pipeline on CIFAR-10 using a ResNet-18 encoder and a two-layer projection head.
- Trained the model with NT-Xent loss for 50 epochs using Adam (learning rate 3×10^{-4}) and batch size 128.
- Evaluated learned representations through linear evaluation, data-scarcity experiments (10%, 50%, and 100% labels), noise robustness tests, and t-SNE visualization.
- Recorded training loss reduction from 5.0557 to 4.3899 and a peak linear evaluation accuracy of 59.93%.

Transformer Reliability Analysis under Domain Shift| Project Link
Transformers, BERT, Hugging Face, NLP — May 2025 – June 2025

- Fine-tuned BERT-base-uncased for sentiment classification on the IMDB dataset using AdamW, batch size 16, learning rate 2×10^{-5}, and maximum sequence length 128.
- Evaluated model reliability under domain shift by testing on both IMDB and SST-2 datasets.
- Achieved accuracy/F1 of 0.8826/0.8843 on IMDB and 0.8589/0.8701 on SST-2, showing a measurable performance drop under distribution shift.
- Analyzed confidence distributions, reliability diagrams, confusion matrices, and misclassified samples to study calibration and error behavior.

GPU Profiling and Optimization for Efficient Deep Learning Training | Project Link
PyTorch, CUDA, GPU Profiling, Performance Engineering — July 2025 – August 2025

- Developed a GPU optimization workflow organized across profiling, training, model, and data components in Python.
- Profiled model training to identify compute and memory bottlenecks.
- Applied mixed precision, batch scaling, and data pipeline optimizations.
- Benchmarked efficiency trade-offs between speed, utilization, and accuracy.

Experience

Research Intern | **Project Link** August 2025 – November 2025

STAR PNT - IIT Tirupati *Tirupati, Andhra Pradesh*

- Built an ML-based framework to correct NavIC/GNSS positioning errors from multipath and NLOS effects.
- Extracted signal features (SNR, elevation, Doppler, residuals) to train models for bias correction.
- Evaluated accuracy improvements through experiments, analyzing error reduction and robustness.

Certifications

CUDA C/C++ and CUDA Python – NVIDIA (October 2024)
Google AI ML Certificate – Google for Developers (September 2025)
Google Cloud Generative AI – Google Cloud (June 2025)
AWS AI Powered Cloud Engineer – AWS Educate (March 2026)

Achievements

Secured All India 4th rank in the 10-day GPU Accelerated Computing and Codeathon in association with NVIDIA.
Selected as a Research Intern for the STAR PNT Challenge conducted by IIT Tirupati among 3000+ participants.
Awarded with Special Prize in a 24-Hour Vibe Coding challenge conducted by Cursors 2025 at ANITS College.
Achieved 48th rank in Engineering and 562nd rank globally on the Unstop platform among 100,000+ participants.
Secured All India 22K Rank among 69K participants in DA Paper for GATE 2026, organised by IIT Guwahati.

Additional Activities

Acted as a Technical Lead for ACM Lendi Chapter 2024.
Paritcipated in Google DevFest 2024 organised by GDG Vizag at Gitam University.
Attended a 3-day Cybersecurity workshop hosted by Cyber Rakshak Team on the topic of Hacking Techniques.
Acted as a Coordinator in the LNIT Summit 2025 - A National event for 1500+ participants.

Declaration

I hereby declare that the information provided above is true and accurate to the best of my knowledge.
I take full responsibility for the authenticity of the details mentioned in this resume.

Place:
Date:

Signature